

Homework for week 4

1. Calculate all the Fourier transforms, in Kreyszig 11.10 Table III. We covered some of these in the practice session, try to do the ones that we didn't without consulting the notes.
2. Calculate Fourier transforms 8, 9, in Kreyszig 11.10 Table I. The table says cosine transform, but you should calculate the complex Fourier transform first, then you can get the cosine transform from the complex one.
3. Calculate all the Fourier transforms listed in the practice 4 problems that we didn't cover, try to do this without consulting the notes.
4. Calculate the Fourier transform of the amplitude modulated signal

$$g(x) = \cos(\omega_0 x) f(x)$$

in terms of $\hat{f}(\omega)$.

Hint: Recall the identity that for $g(x) = f(x + a)$ the Fourier transform is $\hat{g}(\omega) = e^{i\omega a} \hat{f}(\omega)$. Derive the counterpart of this identity for the Fourier transform of $g(x) = e^{i\omega_0 x} f(x)$.